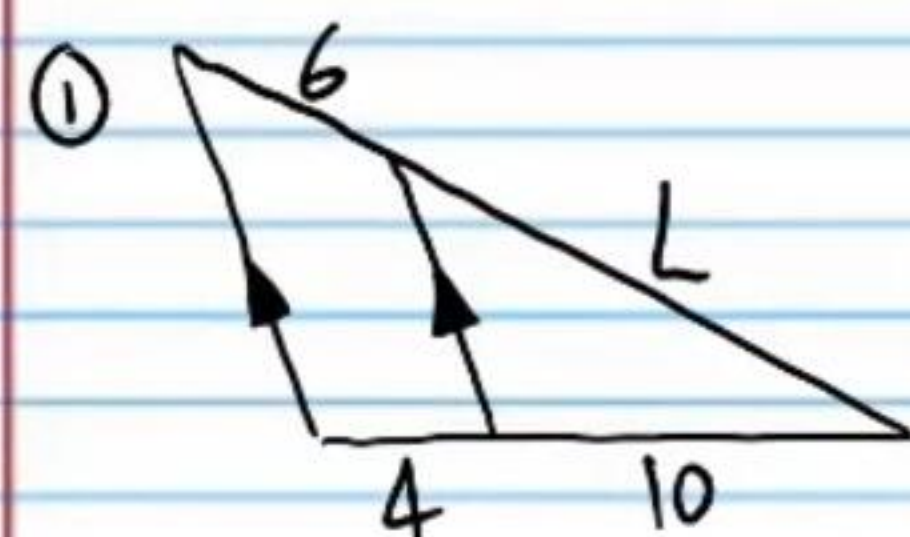


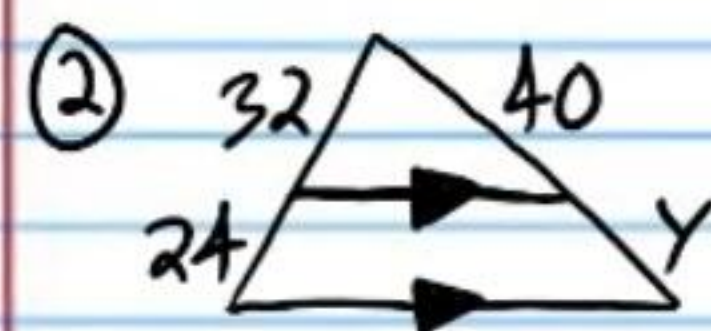
Proportionality Theorems, Transformations, and Perimeter and Area of Similar Polygons (Homework)

Find the unknown values.

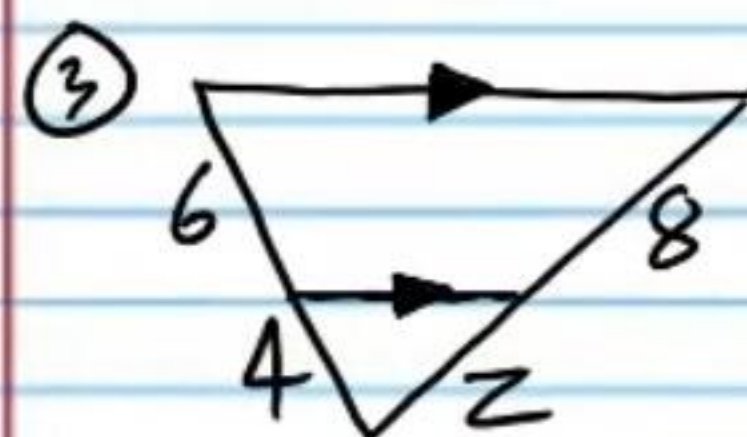


Answers

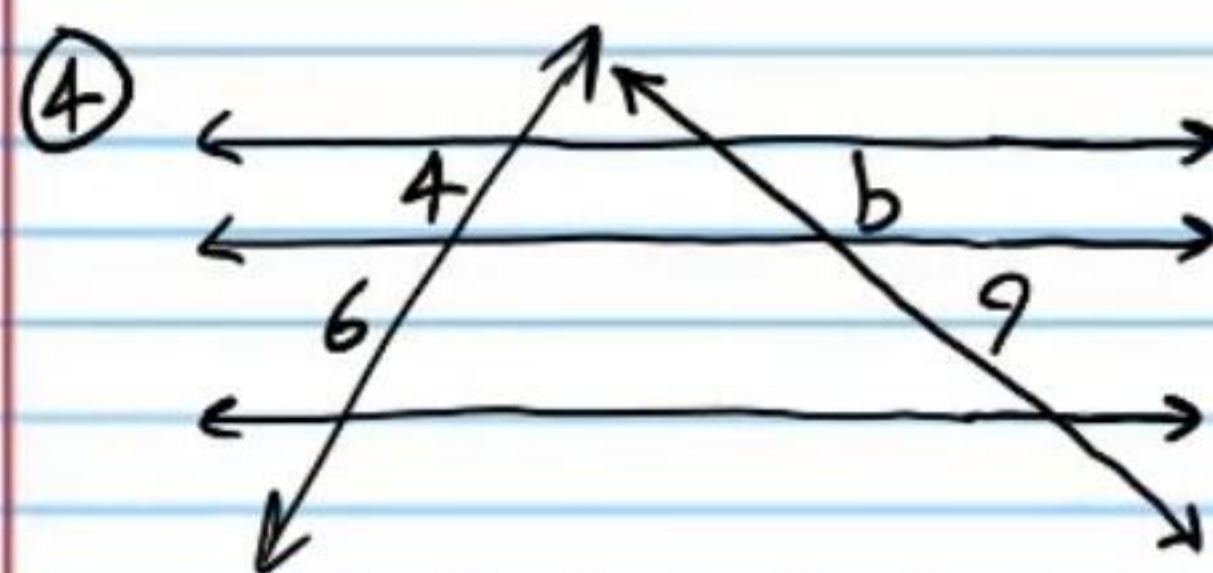
$L = 15$



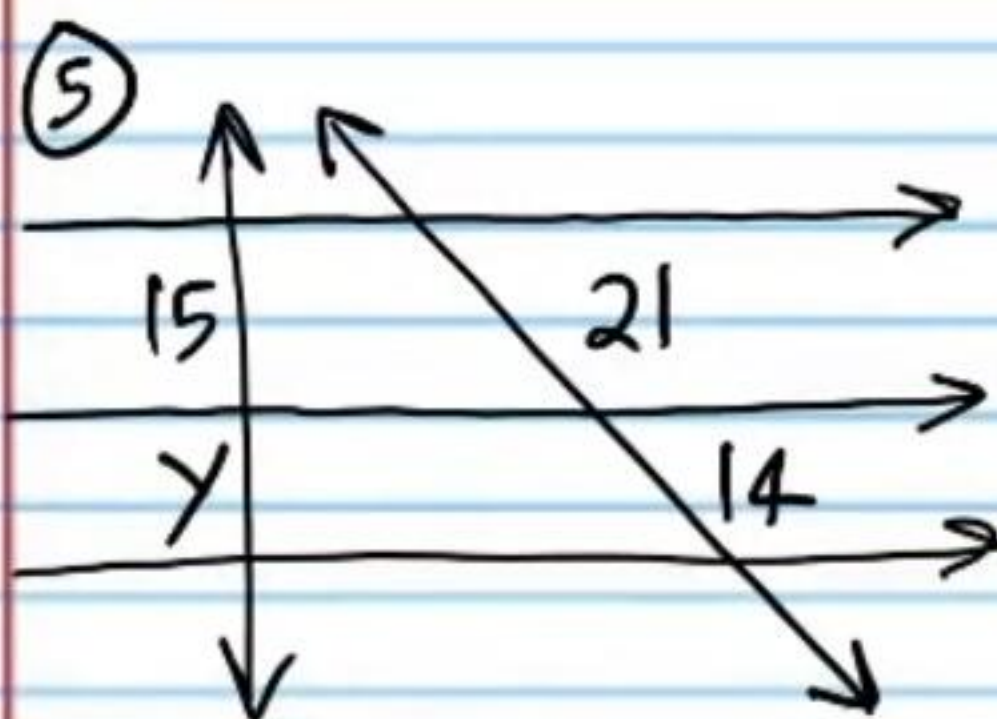
$Y = 30$



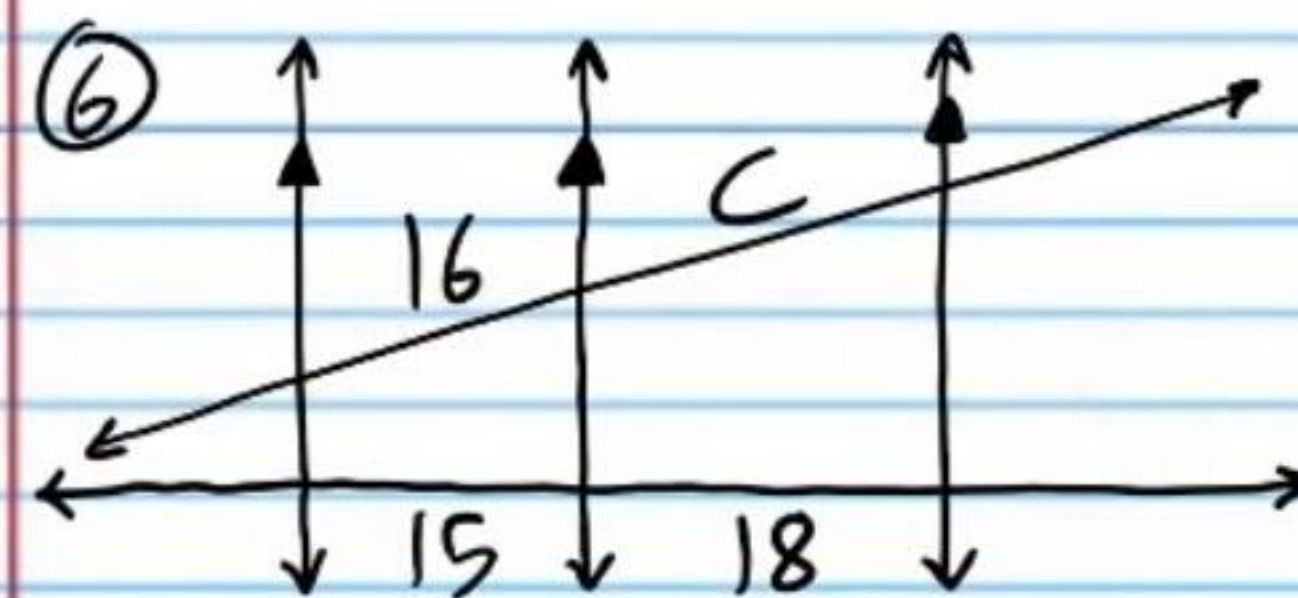
$Z = 5\frac{1}{3}$



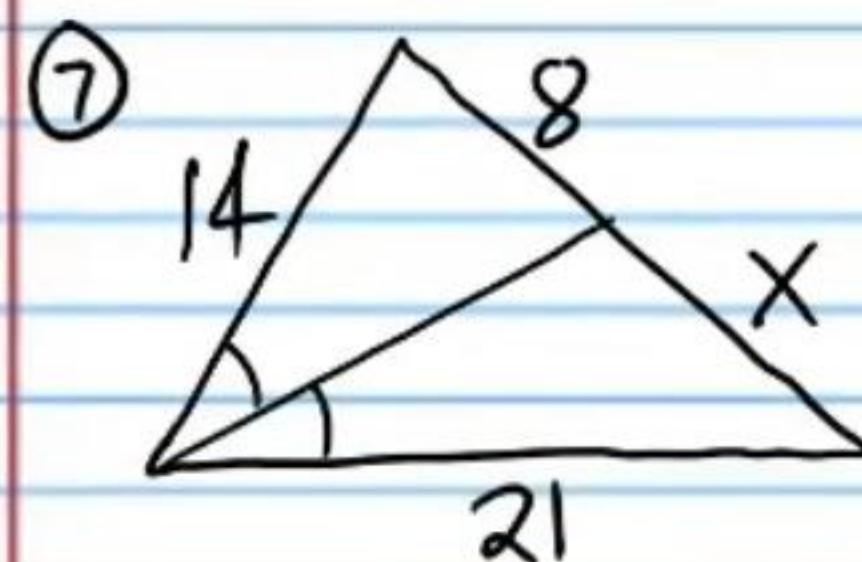
$b = 6$



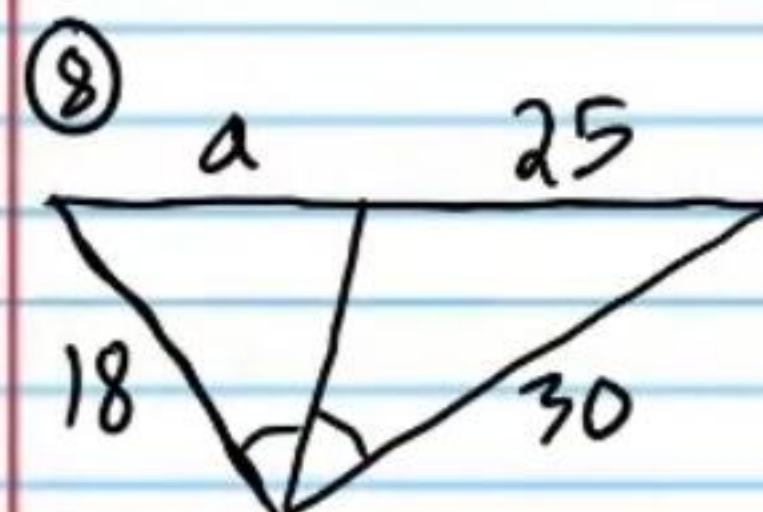
$Y = 10$



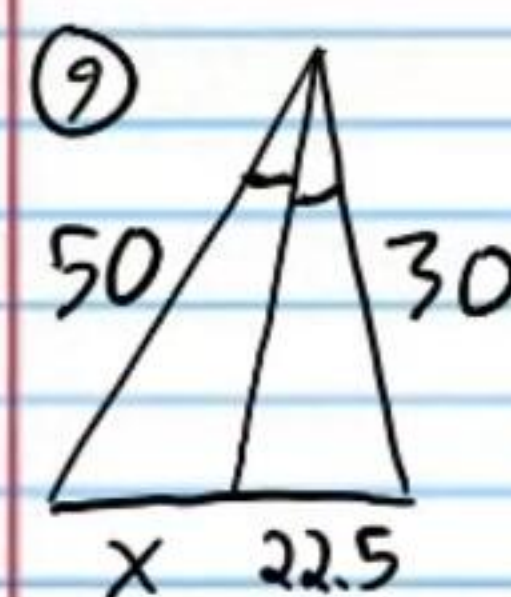
$C = 19.2$



$X = 12$

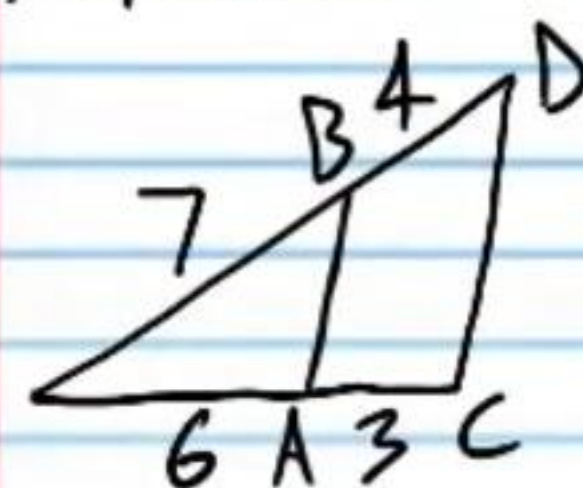


$a = 15$

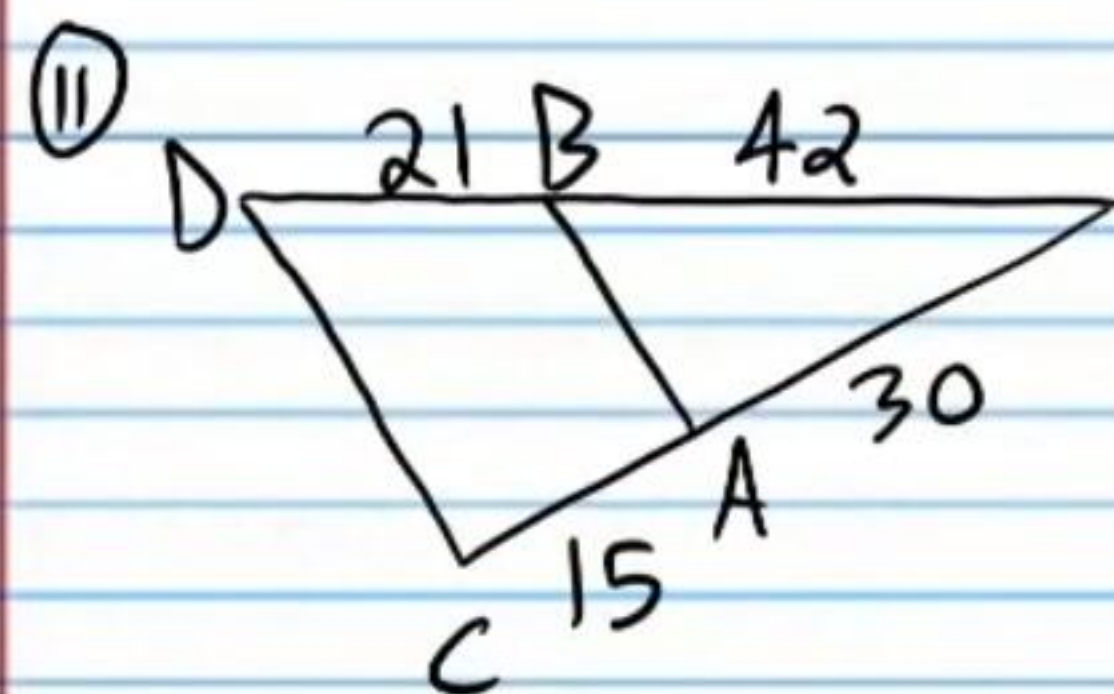


$X = 37.5$

⑩ Is $\overline{AB} \parallel \overline{CD}$? Show the proportions that prove your answers.



No, $\frac{7}{4} \neq \frac{6}{3}$
or $\frac{4}{7} \neq \frac{3}{6}$

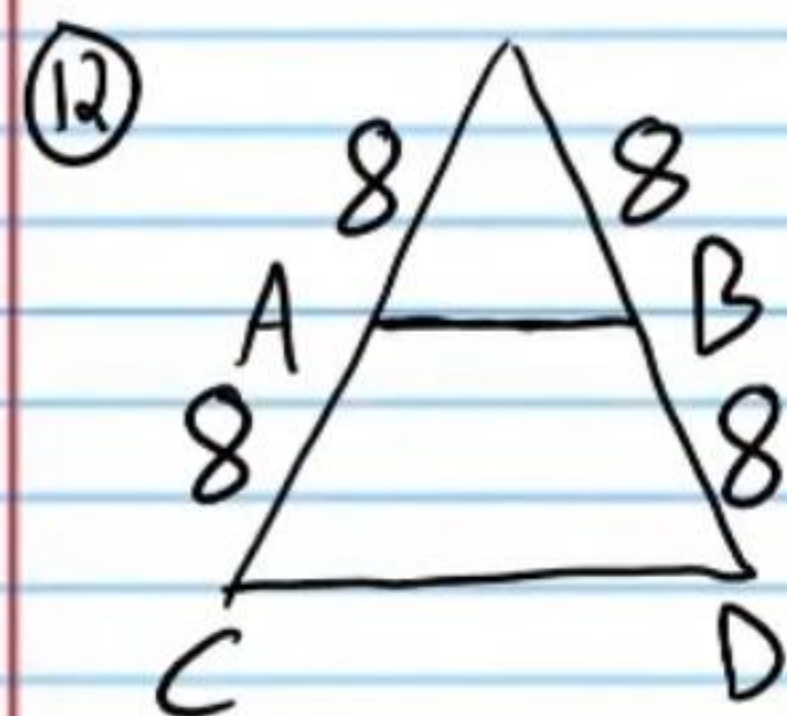


Yes

$$\frac{21}{42} = \frac{15}{30}$$

or

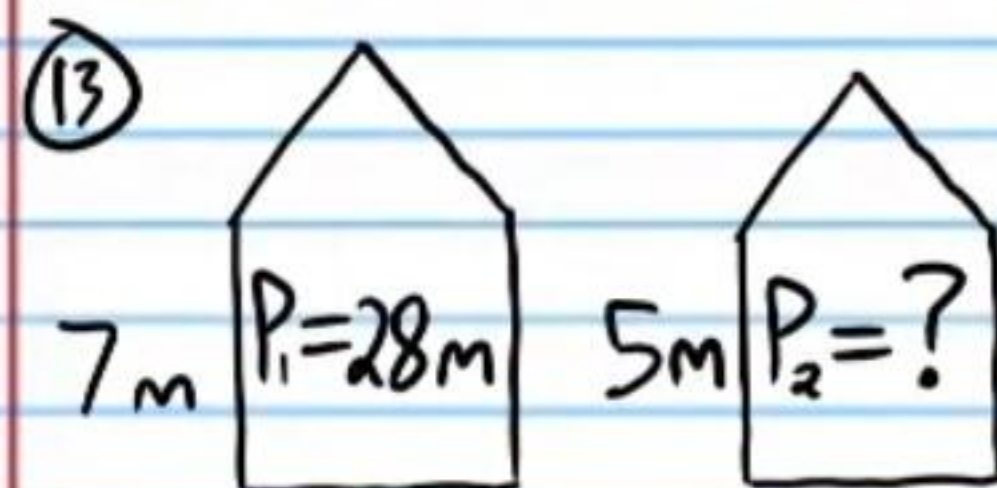
$$\frac{42}{21} = \frac{30}{15}$$



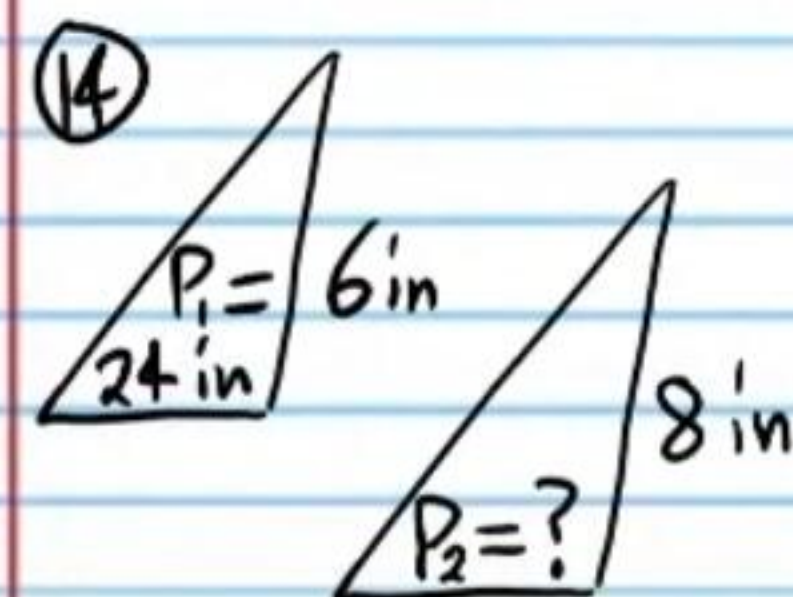
Yes

$$\frac{8}{8} = \frac{8}{8}$$

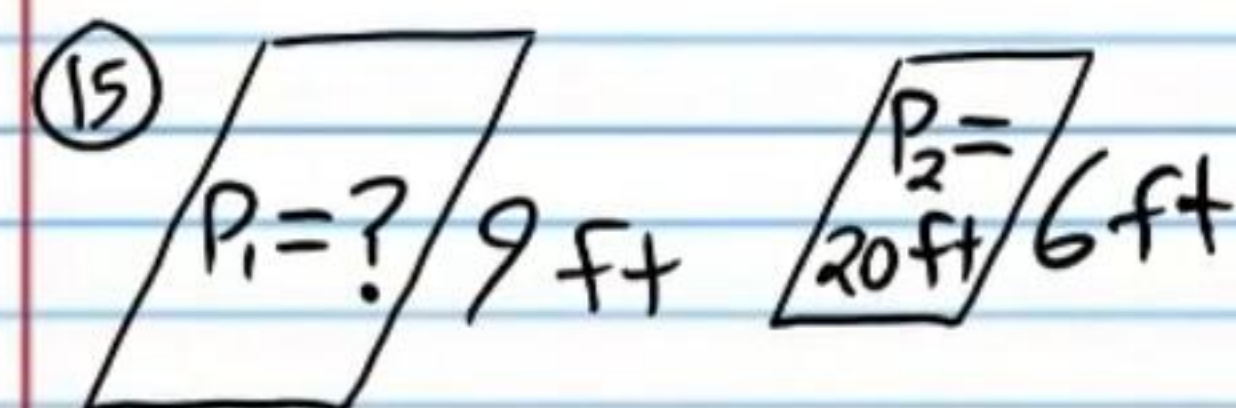
If the following polygons are similar, find the missing perimeters.



20m

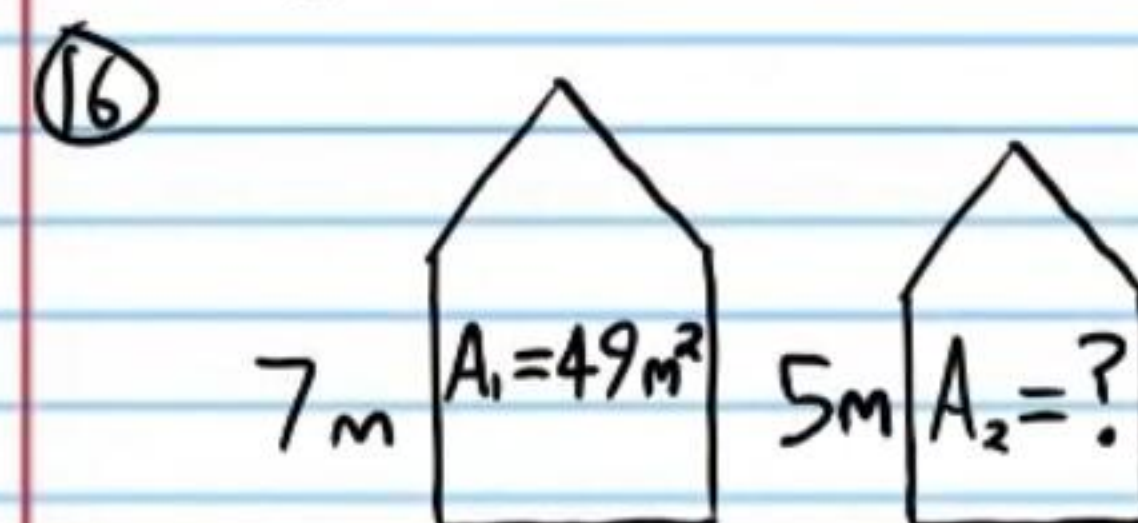


32 in

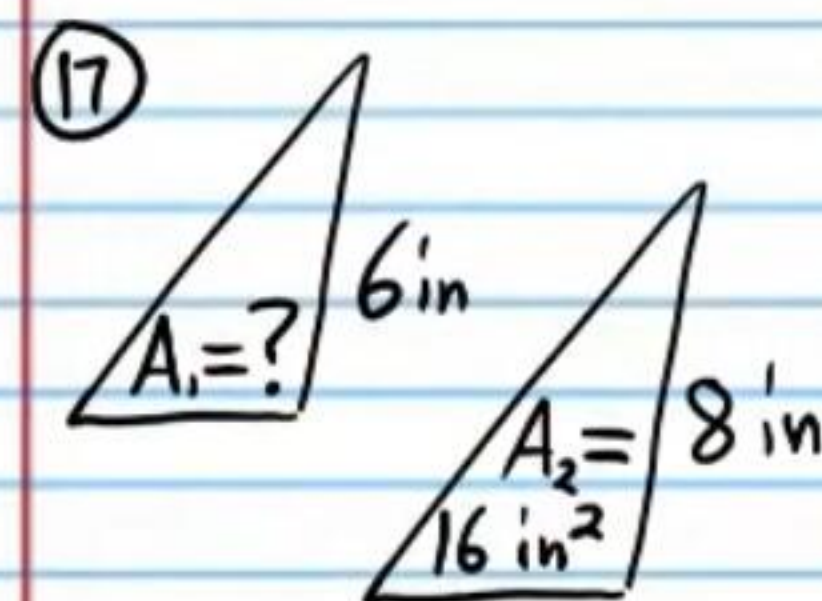


30 ft

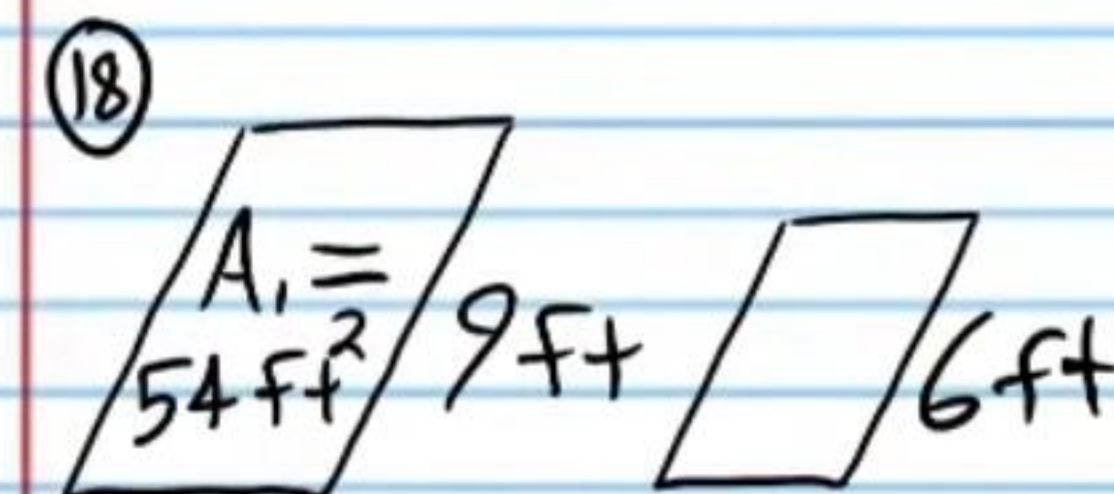
If the following polygons are similar, find the missing areas.



25 m^2



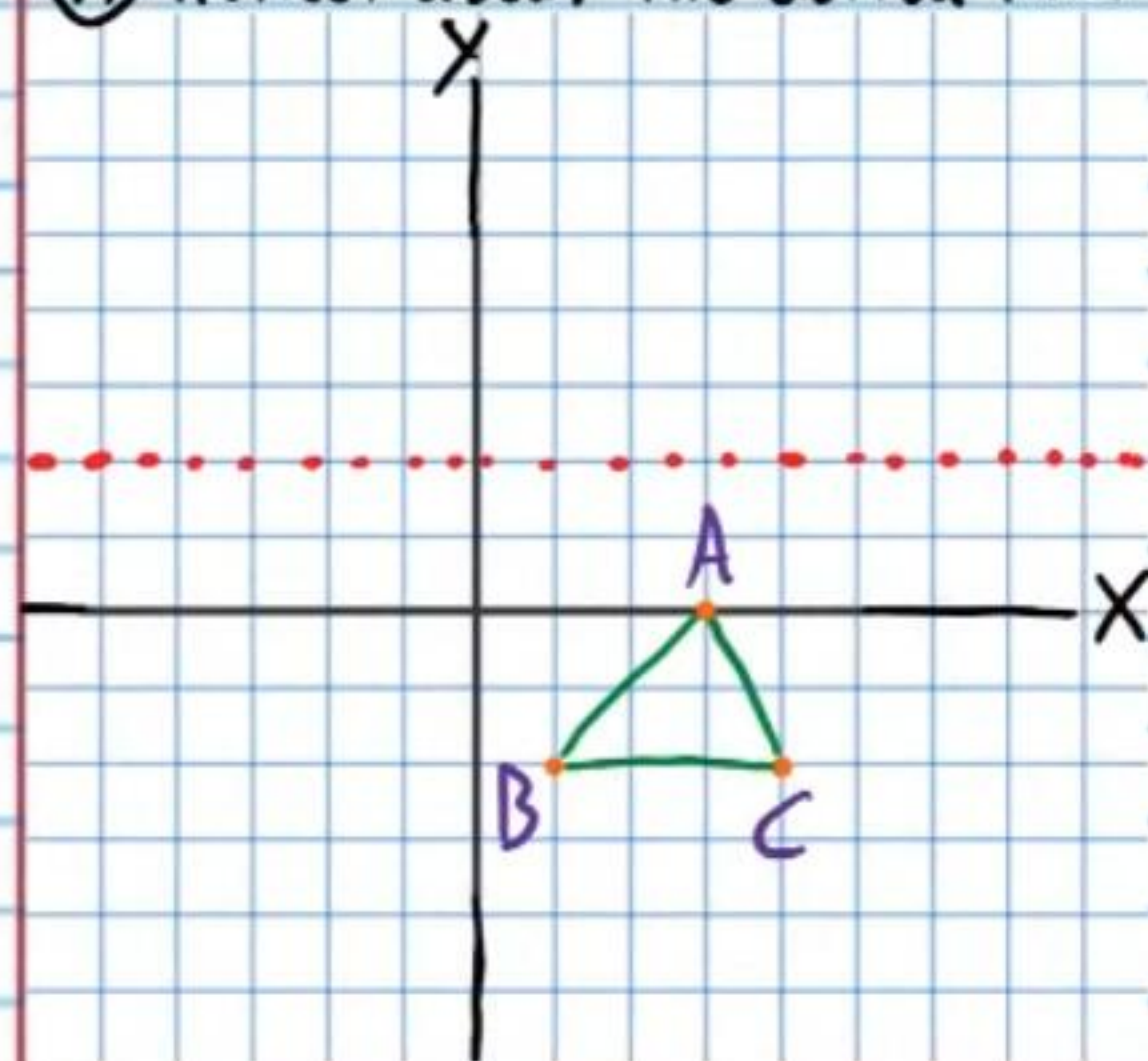
A1 = 9 in^2



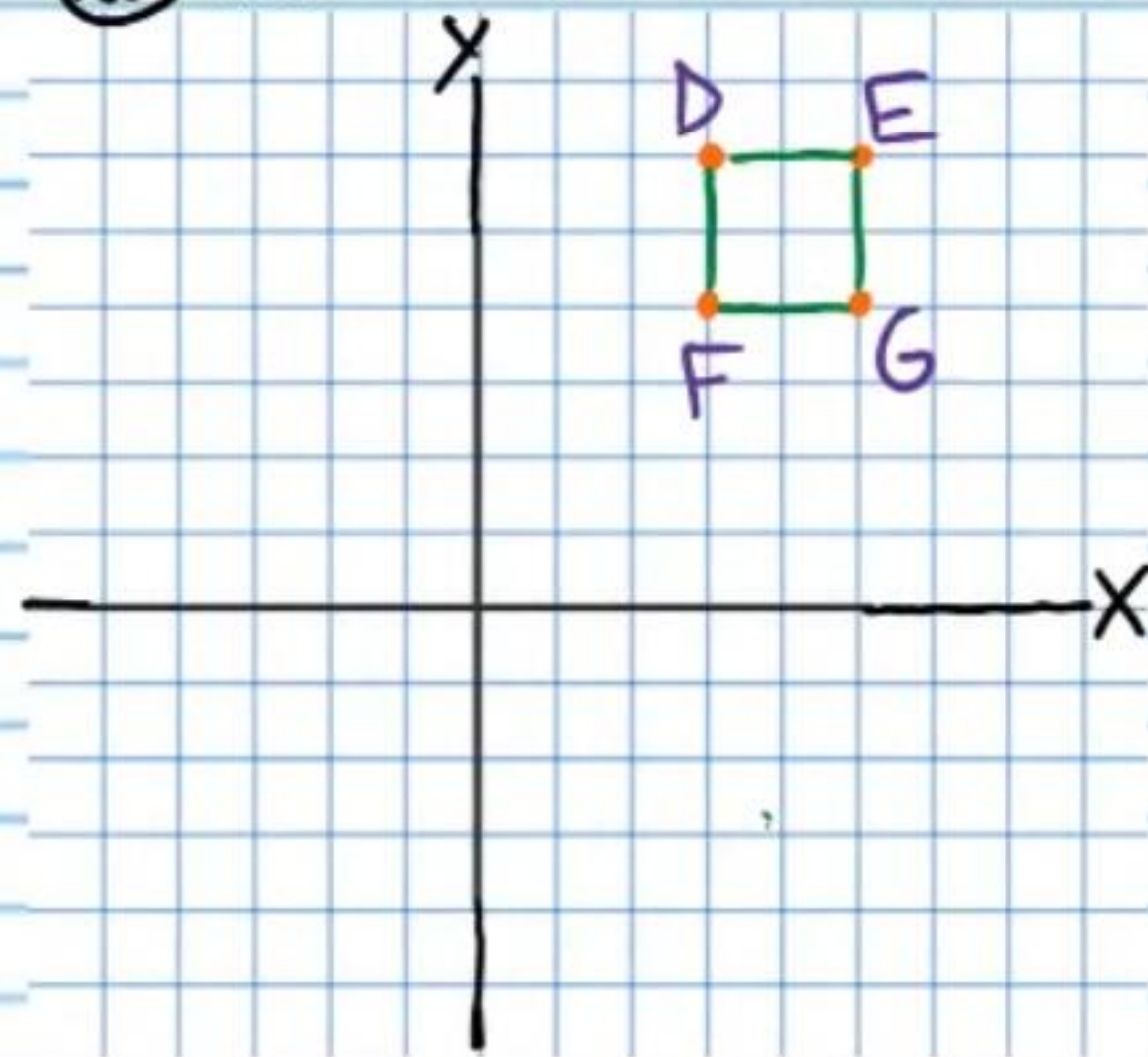
24 ft^2

Given the "pre-image," graph the "image" using the indicated transformation. Answers are on a different page.

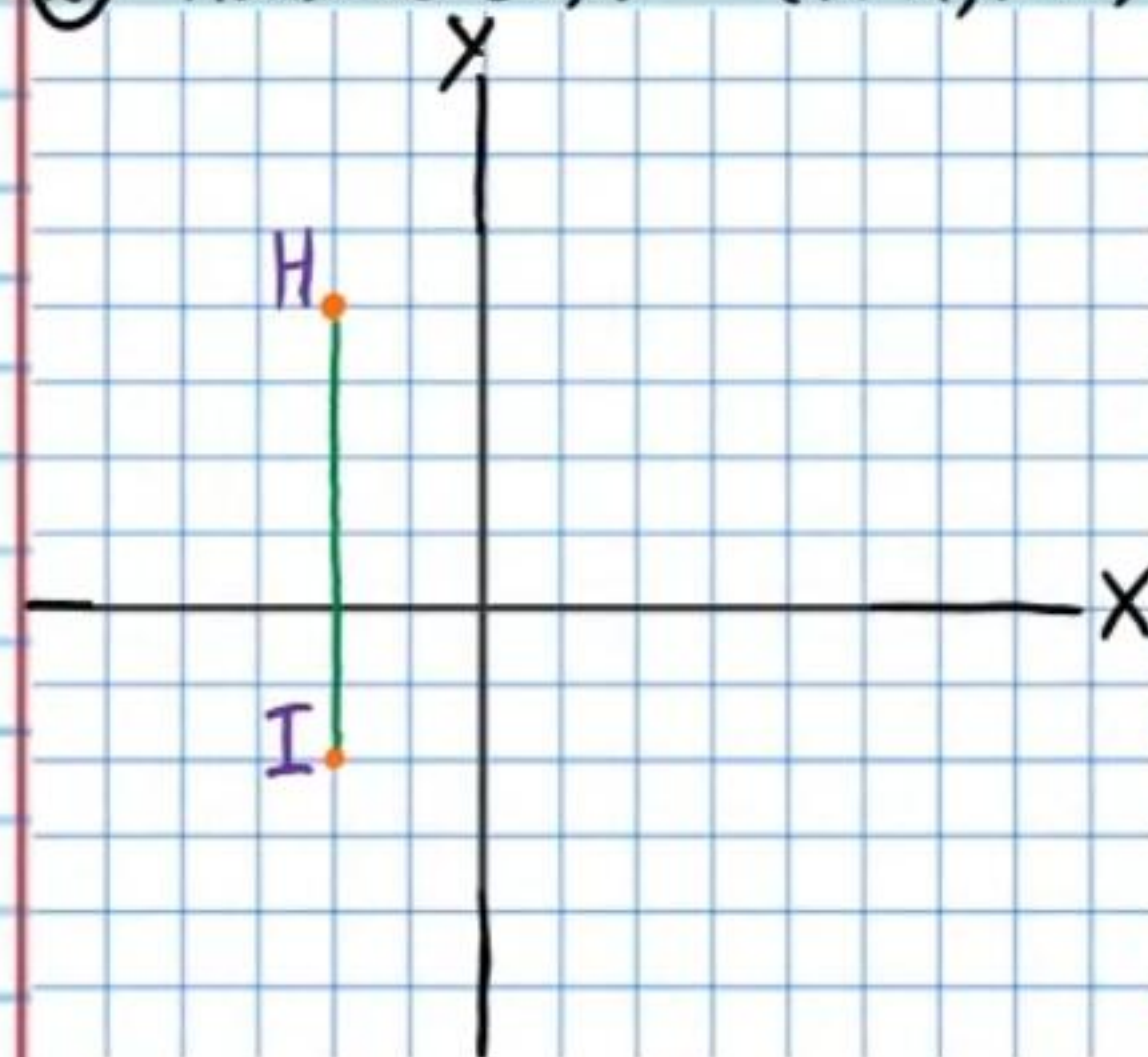
⑲ Reflect about the dotted line.



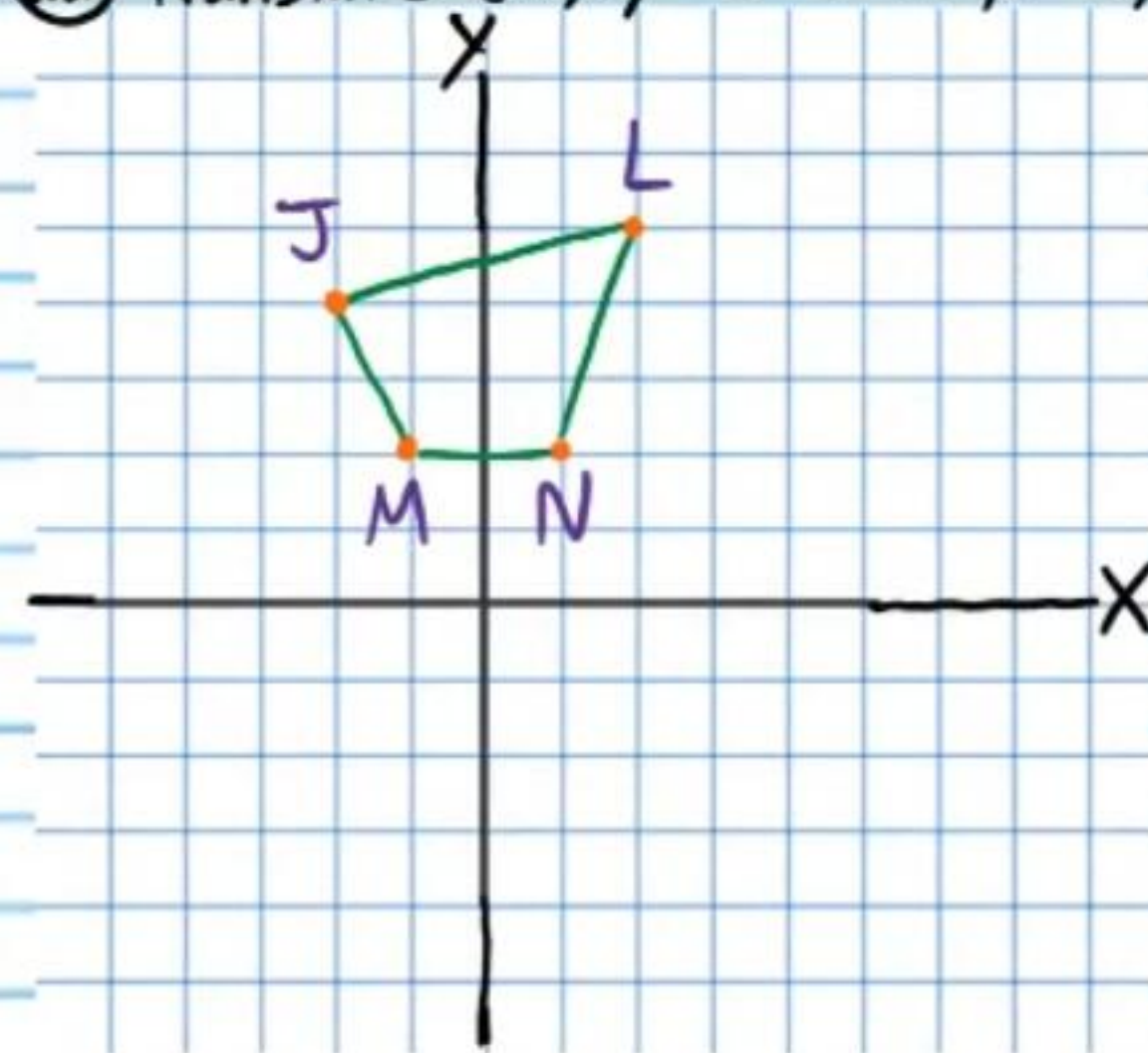
⑳ Reflect about the Y-axis.



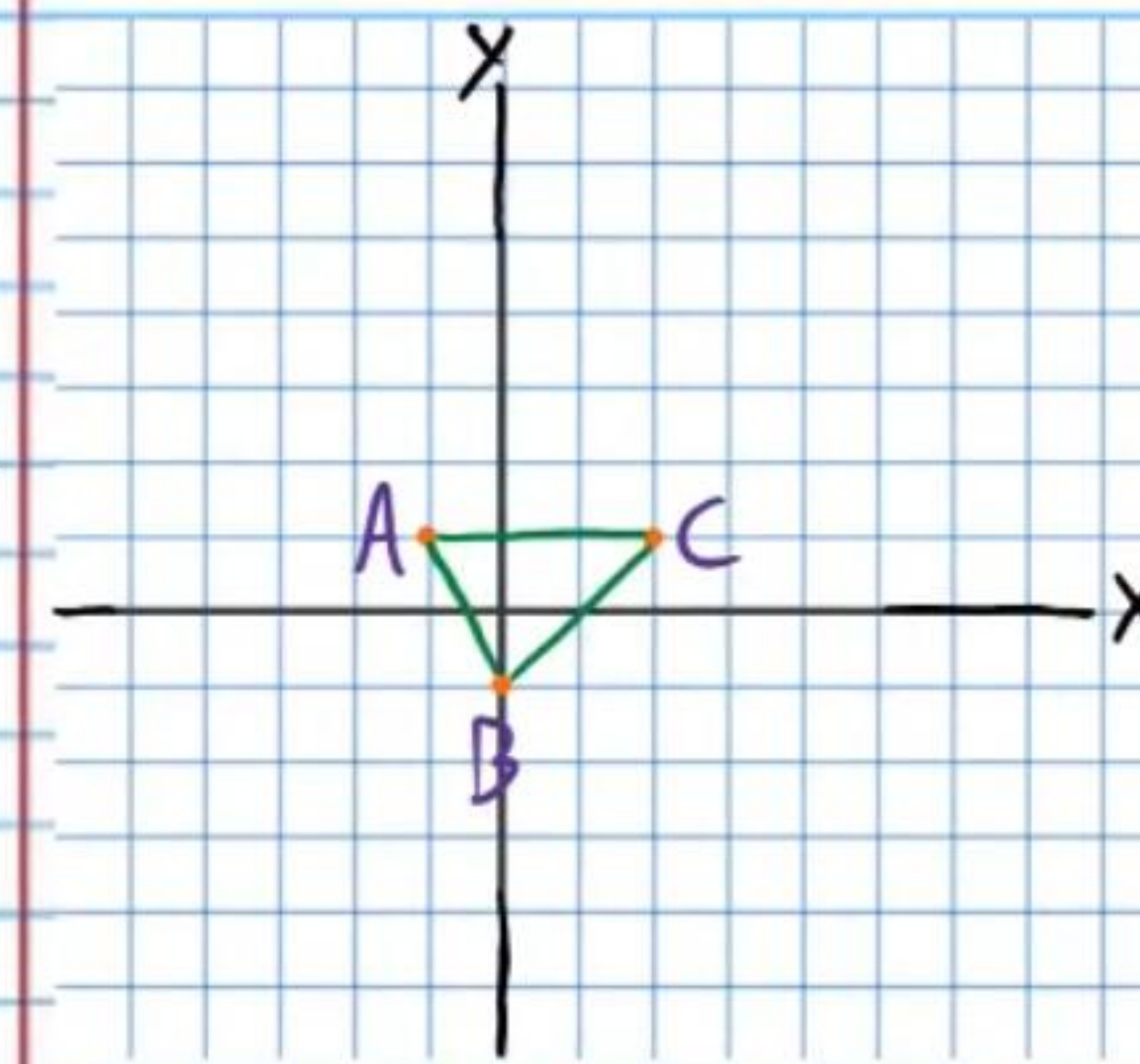
㉑ Translate $(x, y) \rightarrow (x-2, y+1)$



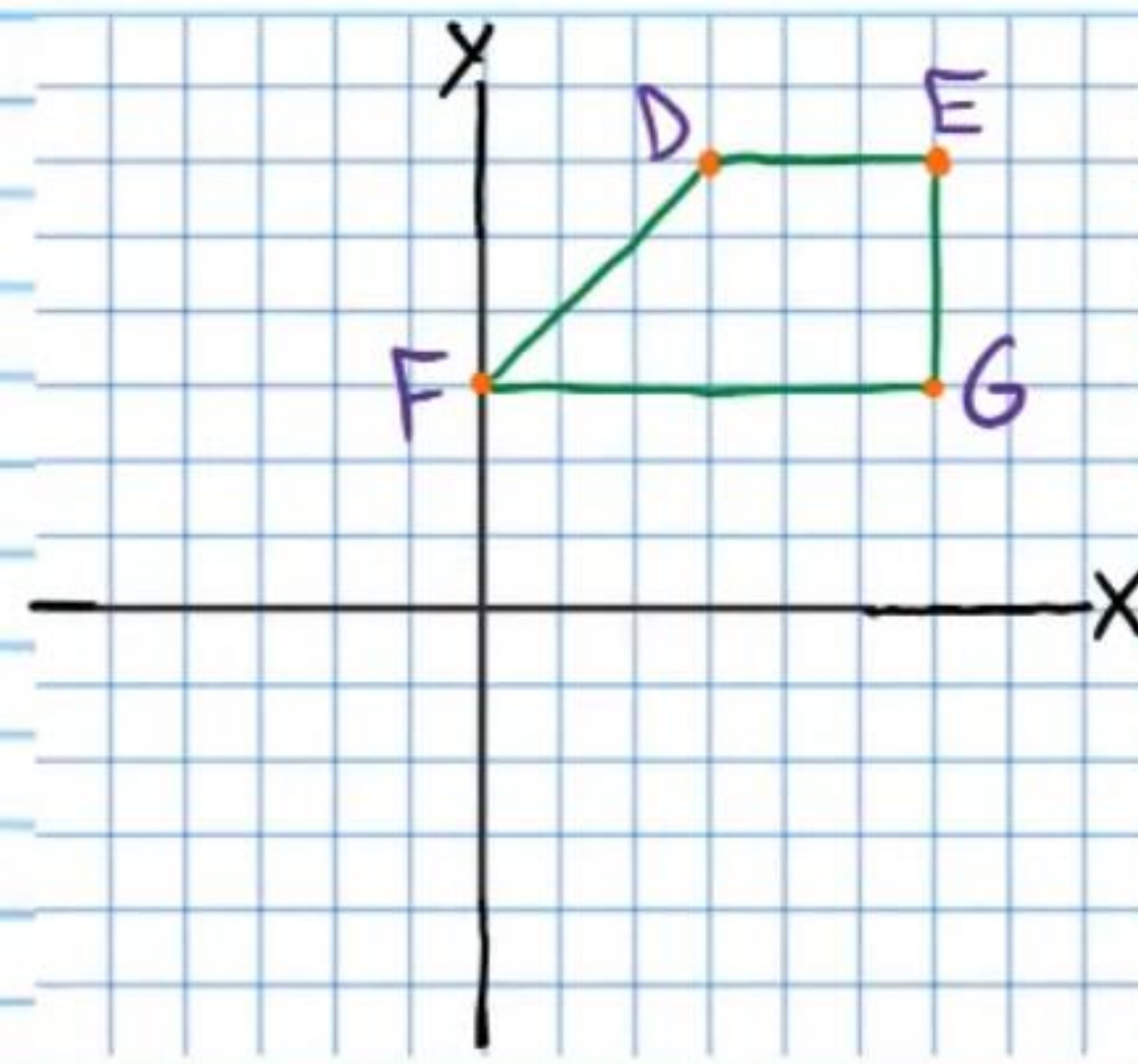
㉒ Translate $(x, y) \rightarrow (x+5, y-5)$



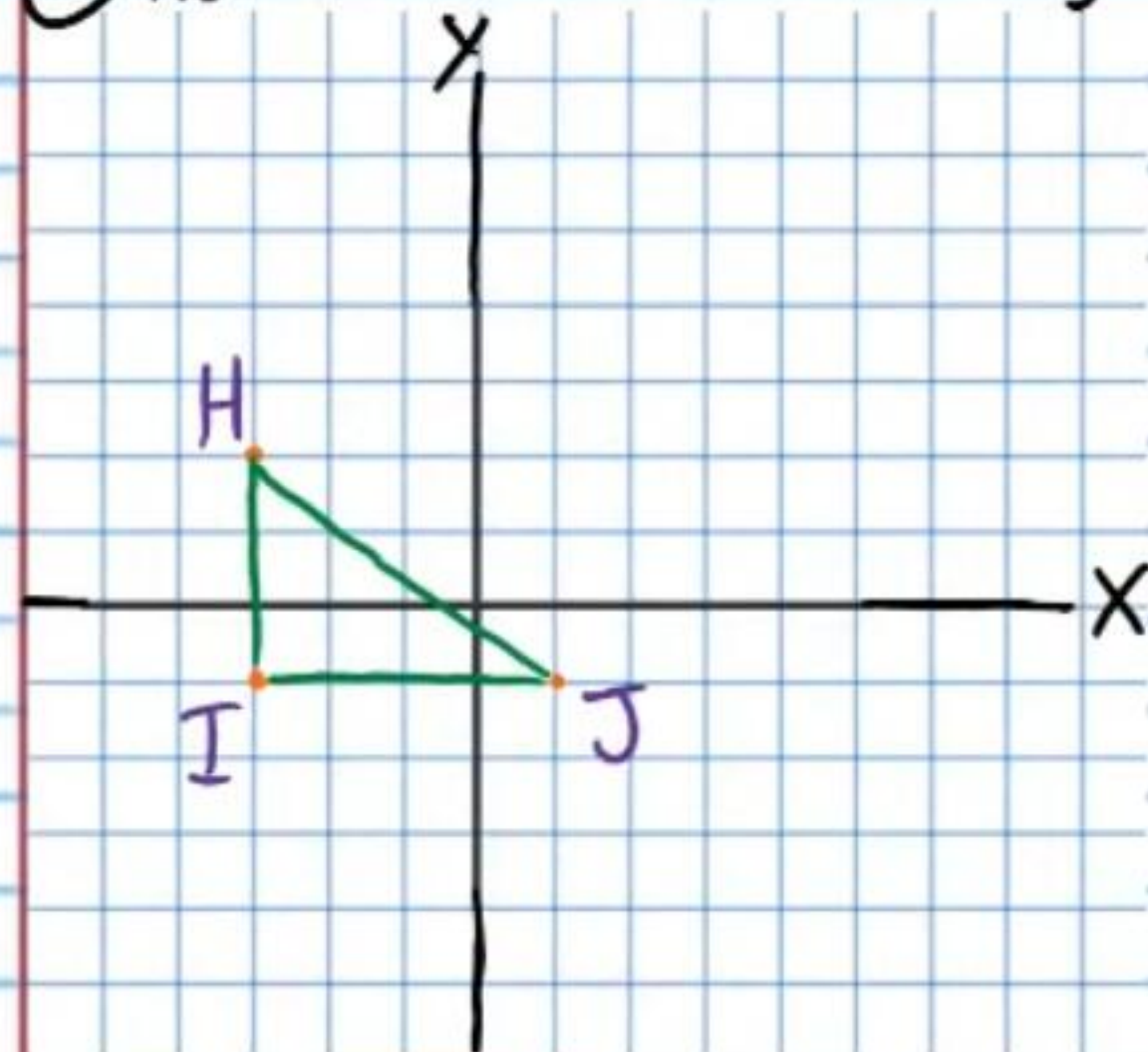
㉓ Dilate about the origin with a scale factor of 4.



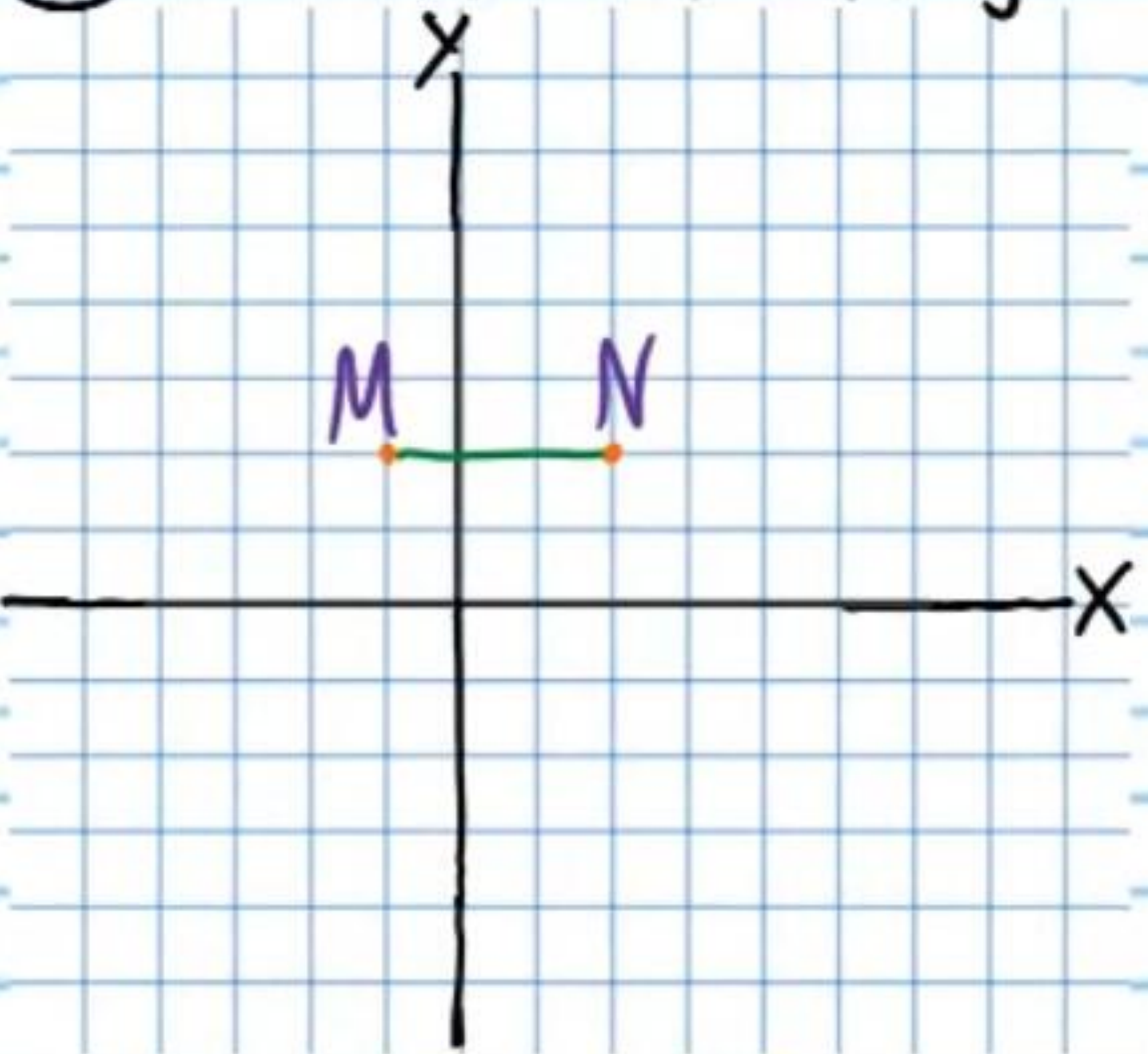
㉔ Dilate about the origin with a scale factor of $\frac{1}{3}$.



㉕ Rotate 180° about the origin.

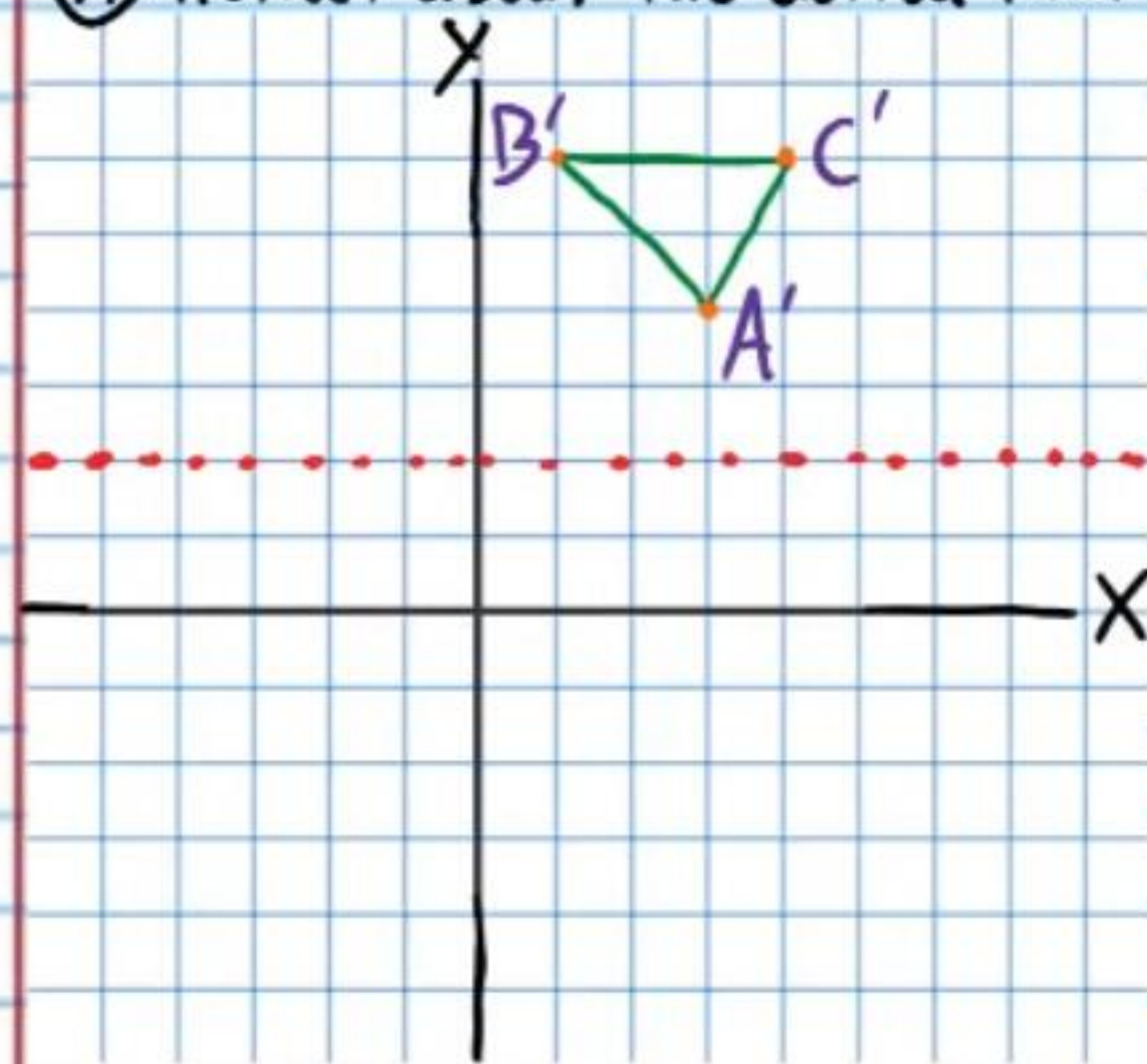


㉖ Rotate -90° about the origin.

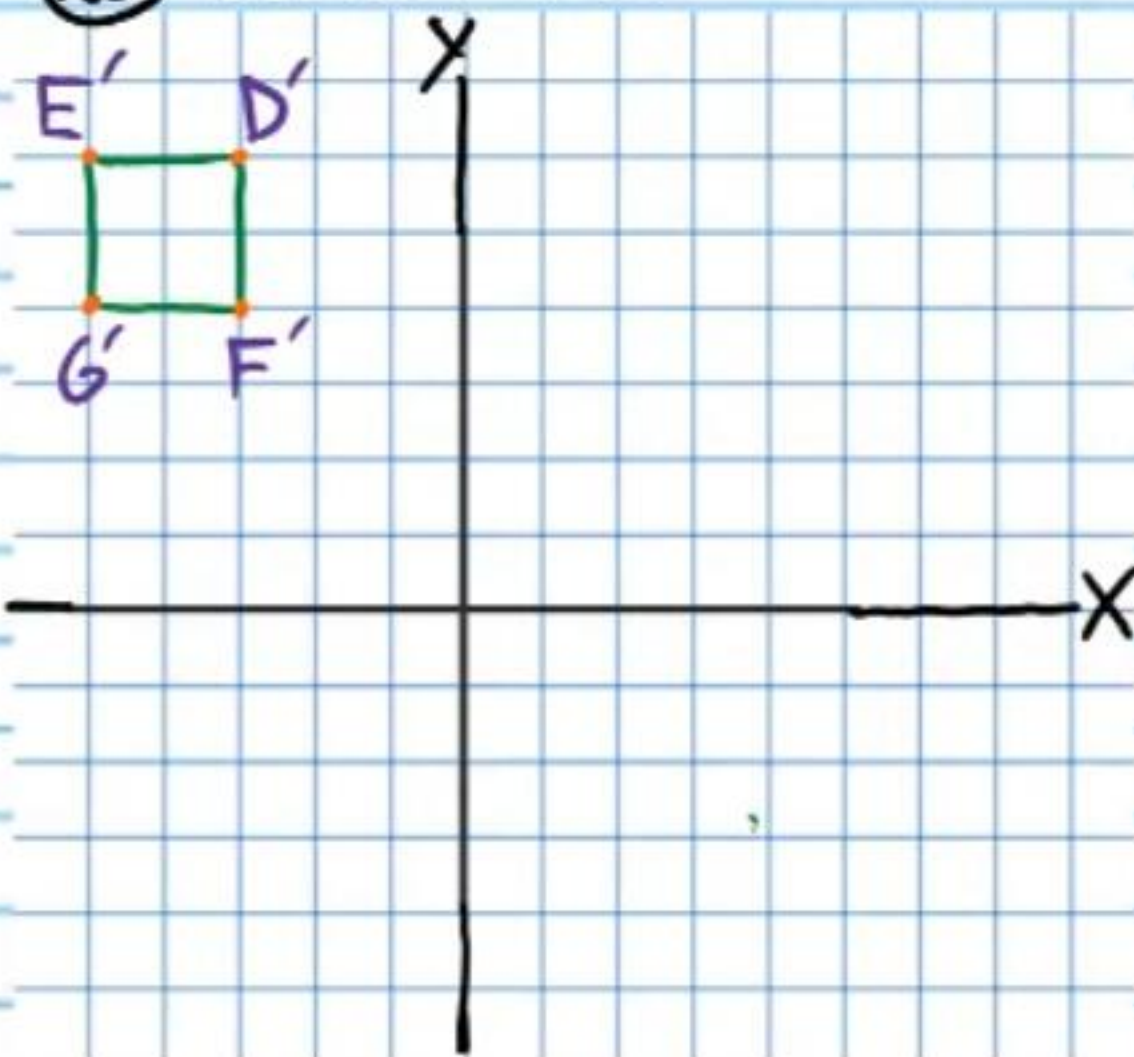


Answers for Problem #'s 19-26

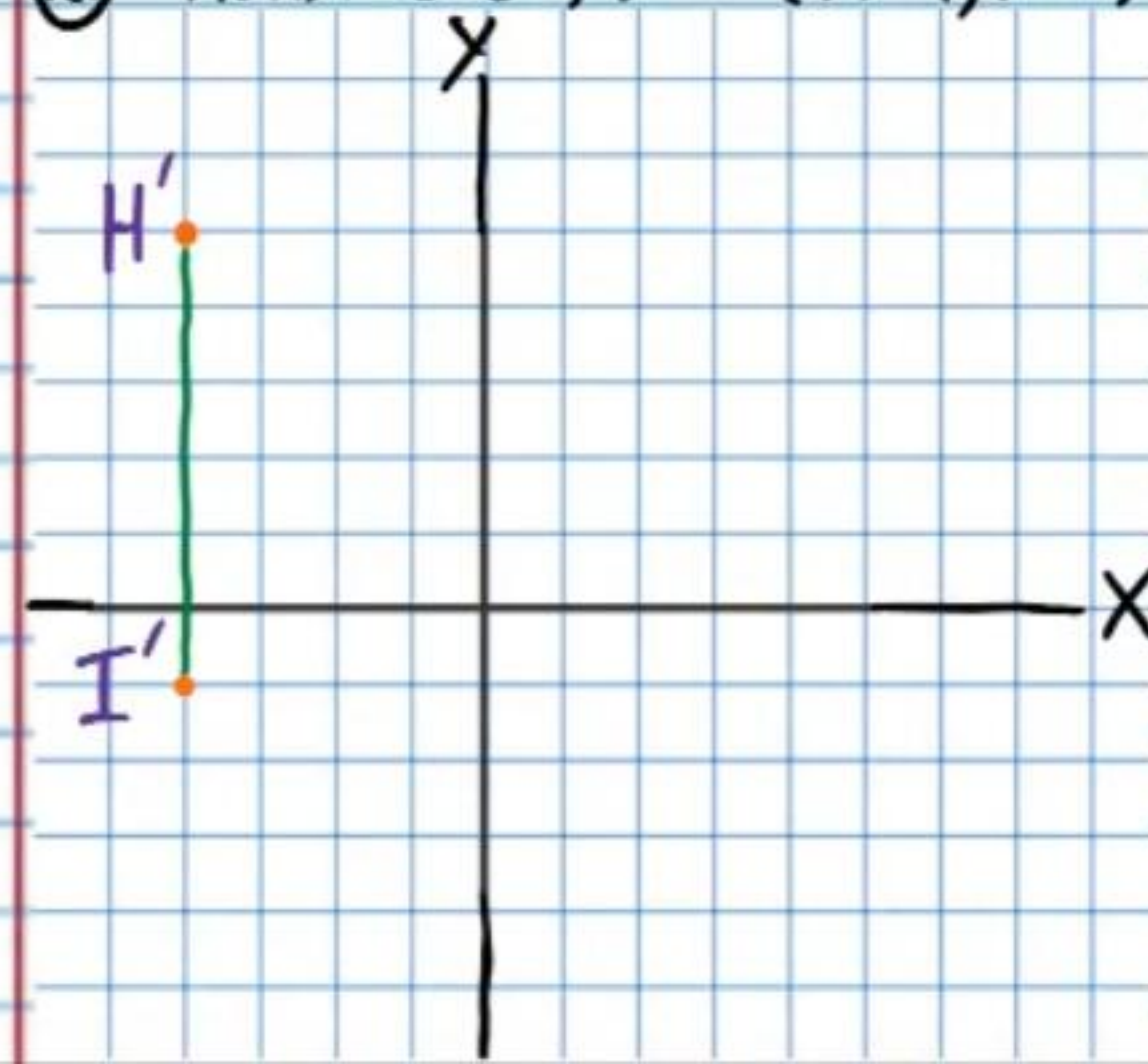
⑪ Reflect about the dotted line.



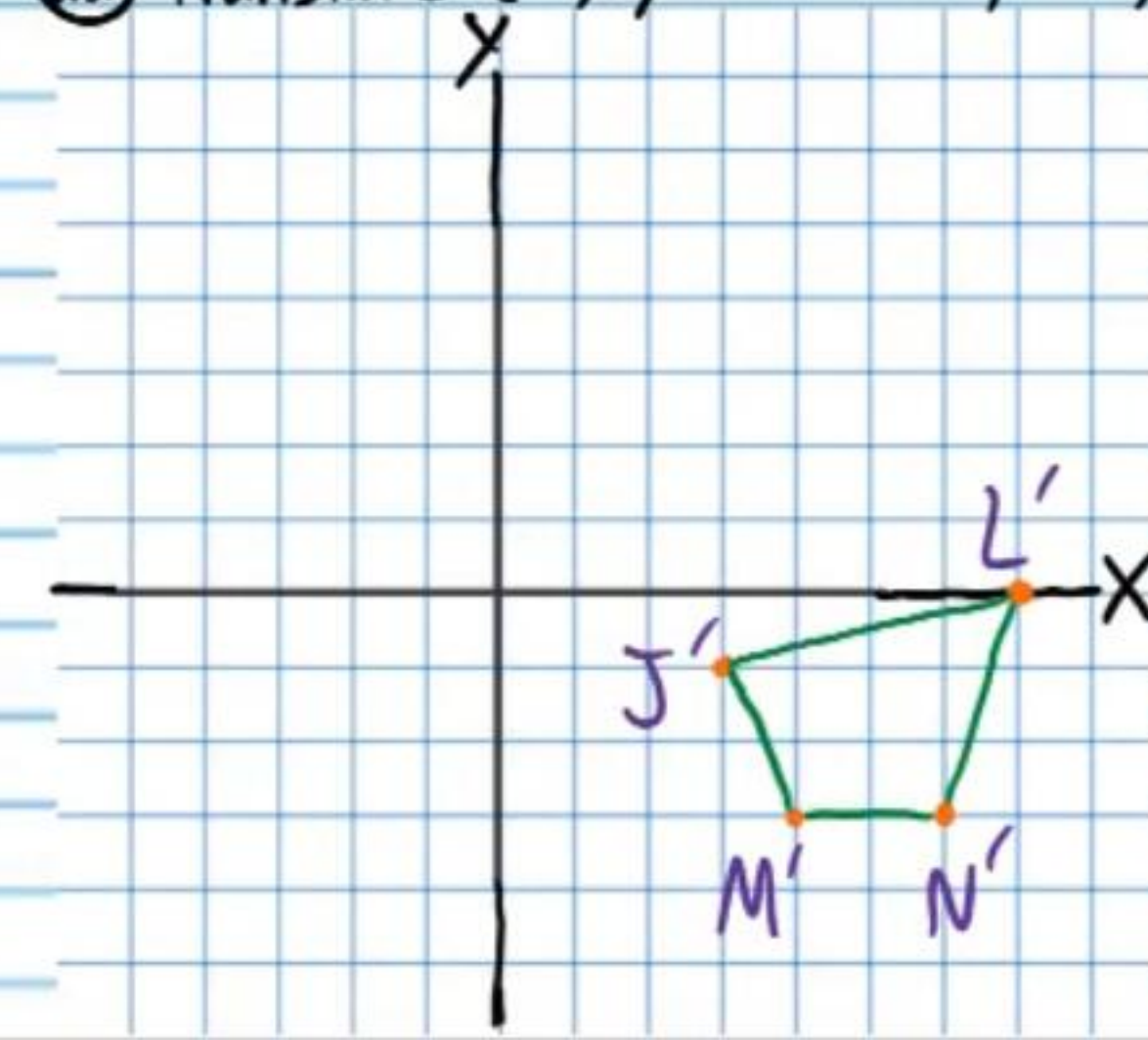
⑫ Reflect about the Y-axis.



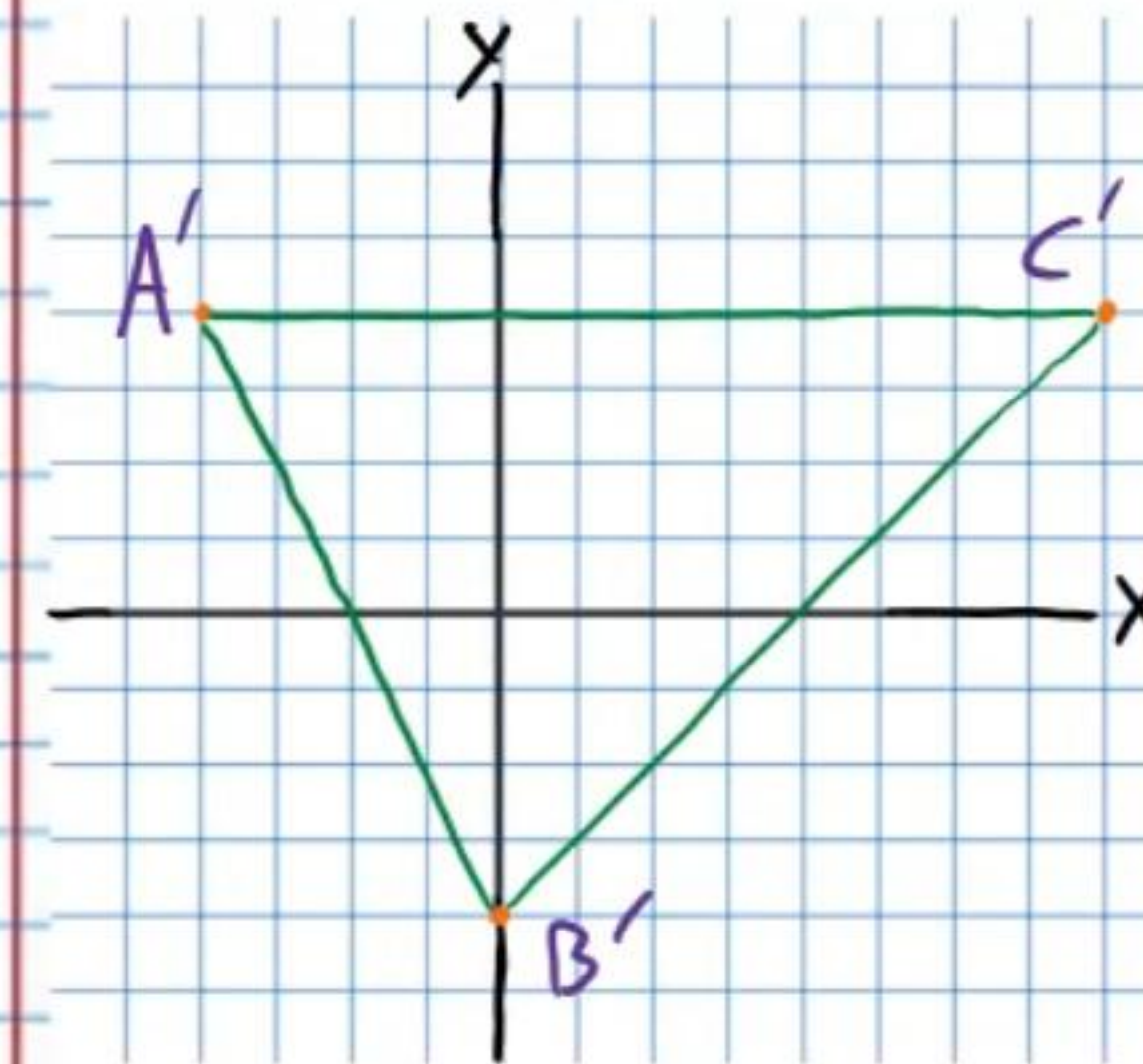
⑬ Translate $(x, y) \rightarrow (x-2, y+1)$



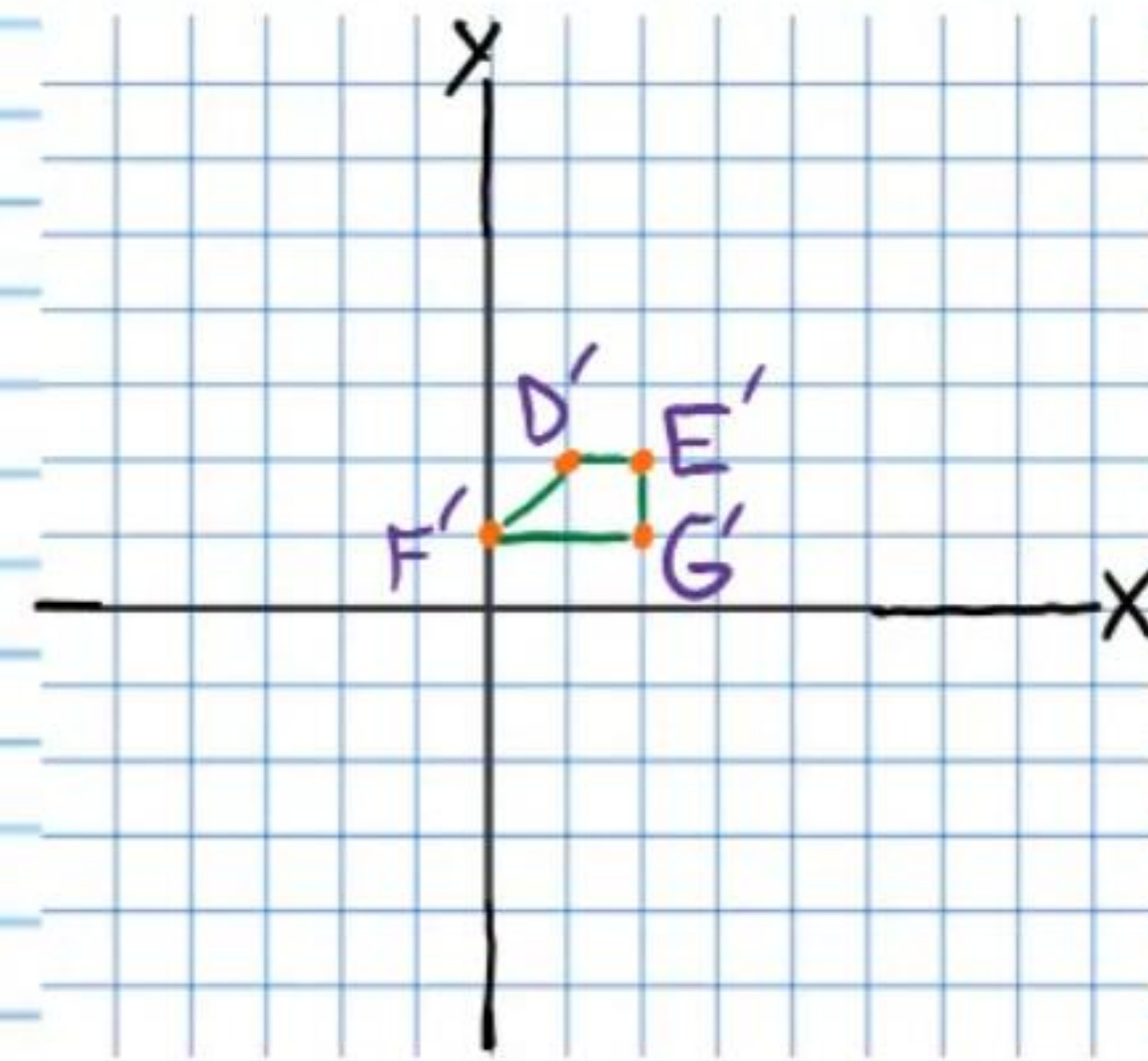
⑭ Translate $(x, y) \rightarrow (x+5, y-5)$



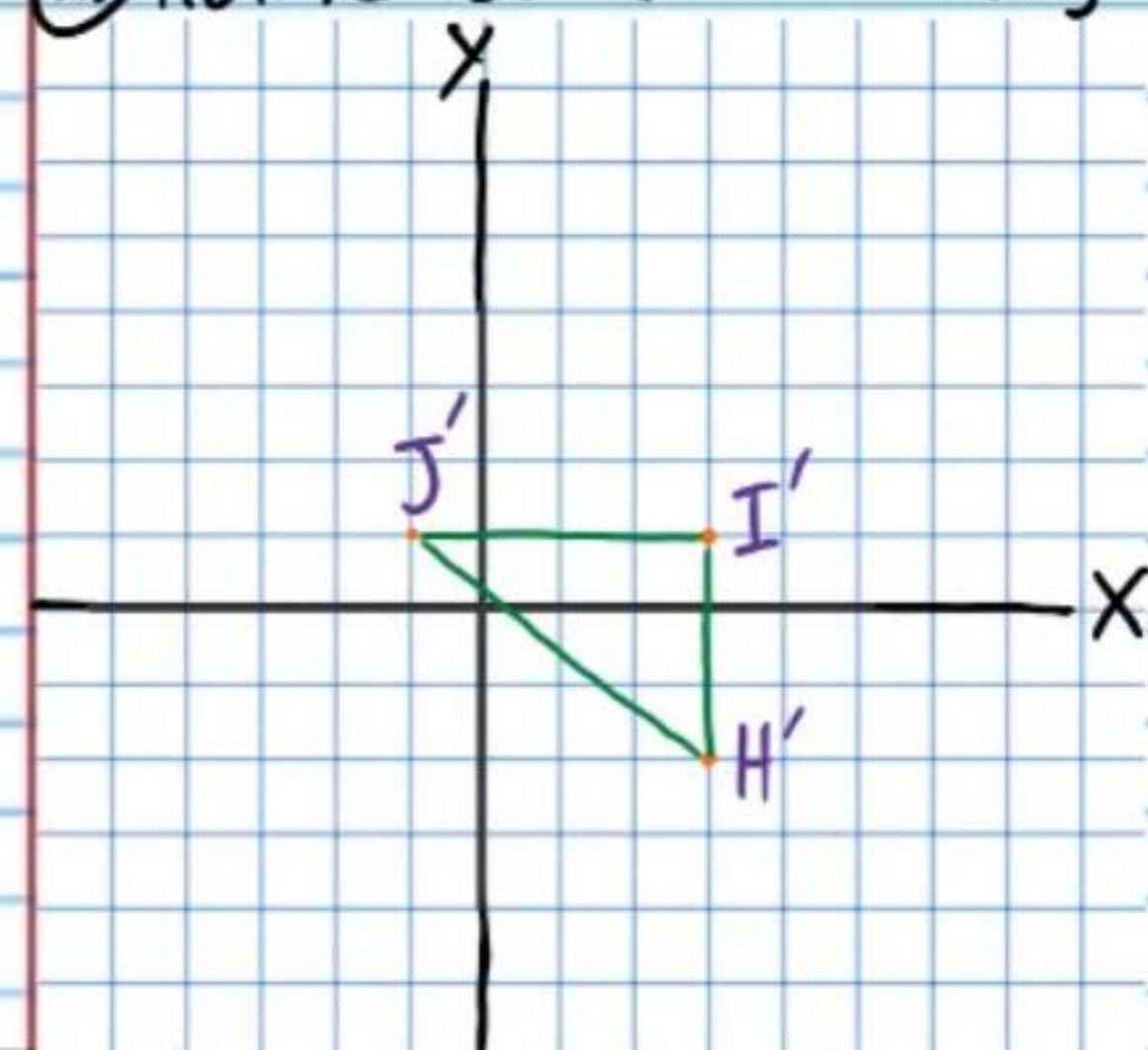
⑮ Dilate about the origin with a scale factor of 4.



⑯ Dilate about the origin with a scale factor of $\frac{1}{3}$.



⑰ Rotate 180° about the origin.



⑱ Rotate -90° about the origin.

